

PAPONTEE JAIDEESAKUNEE

AI Engineer

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OBJECTIVE

AI Engineer specializing in AI agents, voice-driven systems, and applied AI solutions. Combines strong mathematical foundations with hands-on experience designing scalable AI pipelines, integrating full-stack AI systems, and rapidly prototyping intelligent applications for real-world environments.

WORK EXPERIENCE

AI Engineer Intern	True Digital Group	06/2025 - 02/2026
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TrueX AI — AI Agent for Smart Home Control

- Enhanced the TrueX AI Agent with device control features to support end-to-end operations for various smart home appliances and hardware.
- Conducted accuracy benchmarking and evaluation to validate command execution reliability between LLM reasoning and device actions.
- Improved system robustness by testing real-world smart home interaction scenarios.

Noah AI — Next-Generation Voice AI Agent for Multi-Brand Smart Home Control

- Architected and implemented Noah AI, a voice-controlled AI agent integrated with Model Context Protocol (MCP) and Home Assistant.
- Built a low-latency voice orchestration system for multi-brand IoT control.
- Implemented Home Assistant integration enabling control of TrueX smart home devices.
- Converted the Next.js agent chat frontend into a static web application and deployed it within Home Assistant, enabling direct interaction without running a local frontend server.

AI Agent Evaluation Framework

- Designed a Sequential Logic Evaluation framework to validate multi-step AI agent workflows.
- Engineered metrics to verify correct tool invocation order and accurate data passing between tools.

Agentic Vision: MCP for Intelligent Object Localization Pipeline (Find my Item)

- Refactored a legacy object detection pipeline into an MCP-based architecture using DINOv3.
- Developed an embedding-based semantic mapping layer using vector similarity to bridge natural language queries with system labels.
- Designed and conducted benchmark experiments to measure object retrieval precision under controlled environments, evaluating detection accuracy and system reliability.

Voice Infrastructure Optimization

- Migrated from LiveKit (WebRTC) to Pipecat (WebSocket) for agent-human interaction, right-sizing the architecture to reduce costs.
- Simplified the architecture and reduced infrastructure complexity for single-agent voice sessions.

AI Observability & Tracing

- Implemented a tracing and observability system using Opik to monitor complex agent workflows.
- Enabled detailed debugging and performance tracking for production AI systems.

AI Engineer (Part-Time & Intern)

BOTNOI Group

01/2024 - 06/2025

Enterprise RAG System for Insurance Provider

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) ecosystem for enterprise customer service.
- Managed development from research and architecture design to production deployment.

RBAC System with Project-Level Isolation

- Designed and implemented Role-Based Access Control (RBAC) with project-level resource isolation.
- Ensured secure access control across multiple enterprise departments.

B2B API Gateway for RAG Services

- Built a secure API Gateway with API Key authentication enabling external partners to access AI services.
- Developed supporting infrastructure including authentication modules and admin monitoring dashboards.

Backend Microservices Migration

- Contributed to migrating the backend system toward a microservices architecture.
- Improved system scalability and maintainability.

LLM-Based CV Extraction for HR Automation

- Developed an LLM-powered CV extraction system to automate candidate screening and job matching.

Technical Mentorship for Intern Engineers

- Mentored and supervised 5 AI interns, conducted code reviews, and maintained engineering quality standards.

Freelance STEM & AI Tutor

Independent Technical Mentorship

10/2020 - Present

University AI & Engineering Tutoring

- Provided advanced mentoring in Calculus, Machine Learning, Computer Vision, and Deep Learning for university students.

Engineering Faculty Guest Instructor

- Invited by a Faculty of Engineering Calculus professor to deliver preparatory lectures for incoming engineering students.

Competitive School Admission Coaching

- Designed Mathematics and Science programs helping students gain admission into top-tier secondary schools.

EDUCATION

Panyapiwat Institute of Management

2022 - 2026

Bachelor of Engineering (Computer Engineering and Artificial Intelligence)

- GPA: 3.92 (First Class Honors)
- CP Scholarship

SKILLS

AI / Machine Learning: LLMs, Agentic AI, AI Agent Design & Orchestration, LangChain, RAG Architecture, NLP, Embeddings, Vector Search, Semantic Retrieval, AI Agent Evaluation

Computer Vision: OpenCV, Object Detection, Vision Pipeline Integration

Voice AI & Real-Time Systems: Low-latency Audio Streaming, Real-time Voice/Text Communication, WebSocket, WebRTC

Backend & AI Infrastructure: FastAPI, Flask, RESTful APIs, Secure AI Services, WebSocket-based Real-time Systems, Microservices Architecture

Testing & Engineering Practices: pytest, Model Context Protocol (MCP), Structured Logging, Tracing & Observability, Git, Linux

Languages: English (CEFR B1)

ACTIVITIES

Tutor/Teacher

- Faculty Ambassador & IoT Workshop Instructor (September 2024)
- Calculus Foundation for First-Year Engineering Students (June 2024)

Teacher Assistant

- Automated Test Bootcamp 2025 (January 2025)
- Data Engineers Bootcamp 2023 (November 2023)
- MSSQL Database Performance Tuning (October 2023)

CERTIFICATIONS

- Databricks: Data Engineer Bootcamp, Gsoft - January 2026
- Coding with Python Kickstarter, True Digital Academy — January 2025
- Agoda Tech Camp Day, Agoda — November 2024
- Data Analytics Specialize Track, True Digital Academy — March 2024
- Data Analytics Foundation, True Digital Academy — February 2024

REFERENCES

Assoc. Prof. Dr. Parinya Sanguansat, Advisor, Panyapiwat Institute of Management,
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